Overview of Machine Learning and AI

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What is AI?

- AI is the broad field focused on making machines **act intelligently**.

- **Goal**

  To build systems that can *perceive, reason, learn, and act*.

- Includes many subfields:
  - machine learning
  - natural language processing
  - robotics
  - computer vision
  - expert systems
  - and more ……

What is Machine Learning? Big Idea

- Machine Learning (ML) is a subset of AI where systems learn patterns from data, rather than being explicitly programmed.

- ML is when we teach computers to learn from examples, instead of giving them step-by-step instructions.

- **Traditional programming**
  You write rules → computer follows them.

- **Machine learning**
  You give examples → computer figures out the rules itself.

What is Machine Learning? Analogy

- **Like teaching a child**

  If you show a child many pictures of cats and dogs, they eventually learn to tell them apart. You don’t explain every detail (“cats usually have pointy ears, dogs often have floppy ears”), they just learn from examples. That’s how ML works.

- **Like practicing a sport**

  You don’t learn basketball by reading a manual. You practice, get feedback, and improve. Computers do the same with data.

What is Machine Learning? Steps

- **Data**

  The “examples” we feed the computer (photos, text, numbers, sounds, etc.).

- **Training**

  The computer looks for patterns in the data.

- **Model**

  The “recipe” the computer creates to make predictions or decisions.

- **Prediction**

  When shown new data, the computer guesses based on what it learned.

What is Machine Learning? Relatable Examples

- Netflix recommending movies based on what you watched.
- Your email automatically detecting spam.
- Voice assistants (Siri, Alexa) recognizing your speech.
- Self-driving cars learning how to recognize stop signs and pedestrians.

What is Machine Learning?

Math + Data

Types of ML

- Supervised learning
- Unsupervised learning
- Reinforcement learning

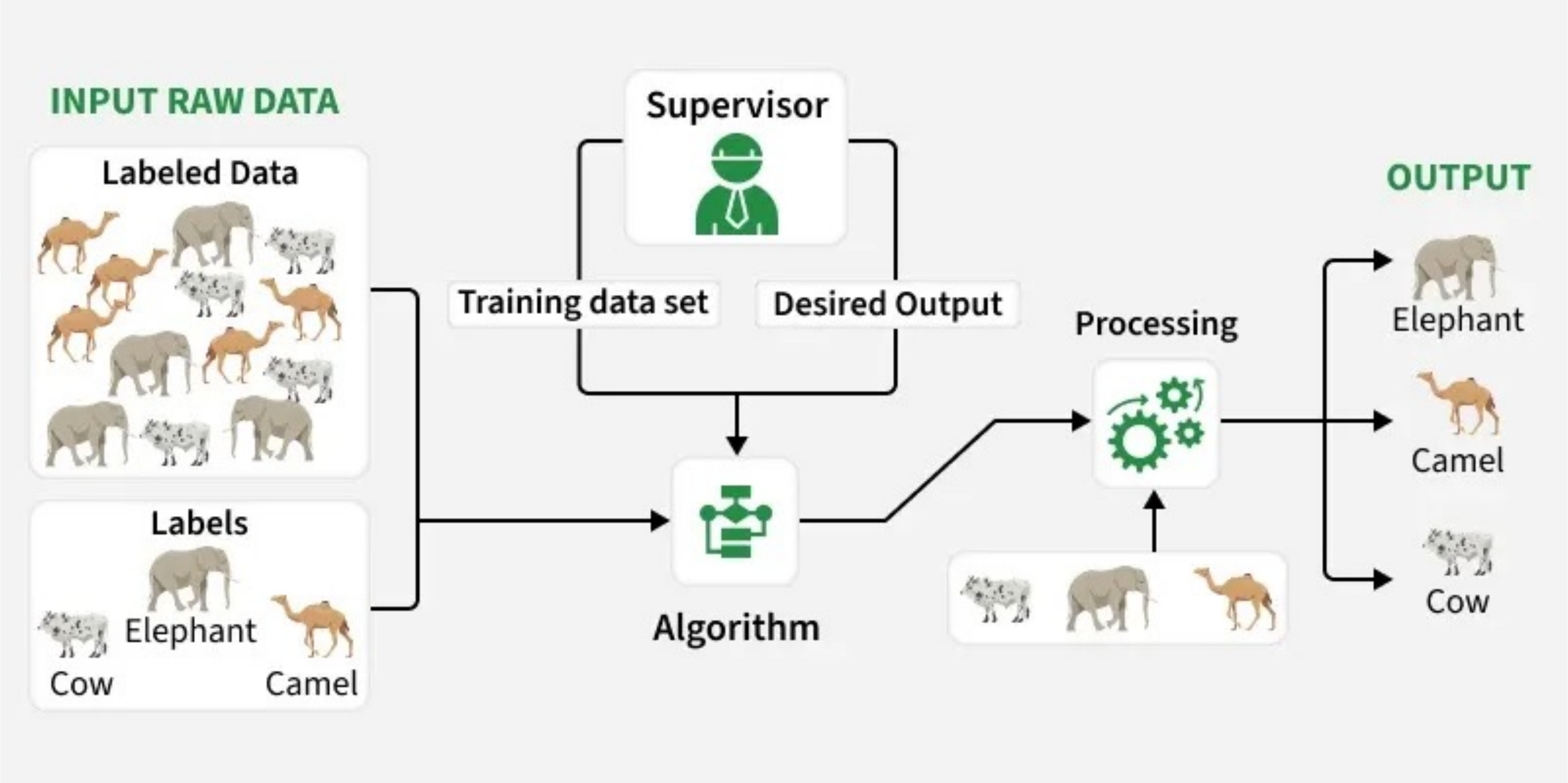
Types of ML: Supervised Learning

- Learn from **labeled examples** (inputs + correct answers).

- Tasks
  - **Classification:** predict discrete labels or categories (spam vs. not spam)
  - **Regression:** predict continuous numerical values (predicting house prices)

- Common methods
  - **Classification:** logistic regression, decision trees & random forests, support vector machines (SVM), neural networks, k-nearest neighbors, Naïve bayes
  - **Regression:** Linear Regression

Types of ML: Supervised Learning



https://www.geeksforgeeks.org/machine-learning/machine-learning/

Types of ML : Unsupervised Learning

- Learn from **unlabeled data**, discovering hidden patterns.

- Tasks
  - **Clustering:** group data points into clusters based on their similarities
  - **Dimensionality reduction:** simplify datasets by reducing the number of features while retaining the most important information
  - **Association rule mining:** find patterns between items in large datasets typically in market basket analysis

- Common methods
  - **Clustering:** K-Means, hierarchical, DBSCAN, expectation-maximization
  - **Dimensionality reduction:** Principal Component Analysis, t-SNE, UMAP
  - **Association Rules:** Apriori algorithm

Types of ML: Reinforcement Learning

- Learn through **trial and error** to maximize rewards.

- Used in robotics, games (AlphaGo), self-driving cars.

- Common methods
  - **Model-based:** Markov decision processes, value iteration algorithm, Monte Carlo tree search
  - **Model-free:** Q-learning, deep Q-Networks (DQN), policy gradient methods

Types of ML: Deep Learning

- A subfield of ML using multi-layered neural networks to model complex systems.
- Excels with very large datasets (images, text, audio).
- Famous architectures:
  - **Convolutional Neural Networks (CNNs):** image recognition
  - **Recurrent Neural Networks (RNNs):** handles sequential data (speech, time-series)
  - **Transformers:** powers natural language processing (GPT, BERT, etc.) and vision tasks
  - **Generative AI:** Creating new data (images, music, text) using models like GANs (Generative Adversarial Networks) or Transformers.
  - **Autoencoders:** Used for unsupervised tasks like denoising or feature learning.

Emerging Trends

- **Transfer Learning**
  Reuses pre-trained models for new tasks (e.g., fine-tuning BERT for specific NLP tasks).
- **Federated Learning**
  Trains models across decentralized devices while preserving privacy.
- **Self-Supervised Learning**
  Learns from unlabeled data by creating pseudo-labels (e.g., contrastive learning).
- **Explainable AI (XAI)**
  Focuses on making ML models interpretable (e.g., SHAP, LIME).
- **AI for Multimodal Learning**
  Combines multiple data types (e.g., text, images) for richer models.

Key Considerations

- **Data Requirements**
  Supervised learning needs labeled data, unsupervised uses unlabeled data, and RL requires an environment.

- **Computational Needs**

  Deep learning often demands GPUs/TPUs, while simpler methods like linear regression are lightweight.

- **Evaluation Metrics**

  Accuracy, precision, recall, F1-score (classification); MSE, RMSE (regression); reward curves (RL).

Challenges and Considerations

- **High dimensionality**

  Data often has many features.

- **Noise & bias**

  ML can learn the wrong patterns if data is biased or messy.

- **Interpretability**

  Many models (e.g., deep learning) are “black boxes.”

- **Ethics & fairness**

  Important to ensure AI is used responsibly.

ML and AI in a Nutshell

- **AI** is the big tent (goal: intelligent machines).
- **ML** is the toolkit (learning from data).
- **Deep Learning** is the heavy-duty engine for big, complex problems.
- Methods can be grouped into **supervised, unsupervised, and reinforcement learning**.

Essential Question

How can machine learning extract useful, reproducible
biological knowledge from data that are
high-dimensional, noisy, heterogeneous, and limited in sample size?

Welcome to:

BINF 6210/8210: Machine Learning for Bioinformatics